Cramer-Rao Lower Bound

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Following the book [Kay], we present some versions of Cramer-Rao Lower Bound theorem.

1 Scalar version

Theorem 1. *Given a probability density function (PDF) $p(x;\theta)$ satisfy the "regularity" condition*

$$E\left[\frac{\partial ln\ p(x,\theta)}{\partial\theta}\right] = 0,\ \forall\theta, \tag{1}$$

where the expectation is taken with respect to $p(x;\theta)$. Then, the variance of any unbiased estimator $\widehat{\theta}$ must not lower than

$$\mathrm{Var}(\widehat{\theta}) \geq \frac{1}{-E\left[\frac{\partial^2 ln\ p(x;\theta)}{\partial\theta^2}\right]}, \tag{2}$$

where the derivative at the true value of θ and the expectation is taken with respect to $p(x;\theta)$. Furthermore, the equality occurs for a specific unbiased estimator $\widehat{\theta} = g(x)$ if and only if

$$\frac{\partial ln\ p(x;\theta)}{\partial\theta} = I(\theta)(g(x)-\theta),\ \forall\theta, \tag{3}$$

for some functions g and I. $\widehat{\theta} = g(x)$ is called MVU estimator and the minimum variance is $\dfrac{1}{I(\theta)}$.

Proof. Let $g(\theta) = \alpha$. We consider all of unbiased estimator $\widehat{\alpha}$ with $E(\widehat{\alpha}) = \alpha = g(\theta)$.
We consider the regularity condition

$$E\left(\frac{\partial ln\ p(x;\theta)}{\partial\theta}\right) = 0.$$

$$\begin{aligned}
\int \frac{\partial ln\ p(x;\theta)}{\partial\theta} p(x;\theta)dx &= \int \frac{\partial p(x;\theta)}{\partial\theta}dx \\
&= \frac{\partial}{\partial\theta}\int p(x;\theta)dx \\
&= \frac{\partial 1}{\partial\theta} \\
&= 0.
\end{aligned}$$

This condition is simply just the interchange property between integration and derivative. Then, we can check the regularity condition by verifying this property.
From the unbiased estimator $\widehat{\alpha}$,

$$\int \widehat{\alpha} p(x;\theta) = g(\theta). \tag{4}$$

Differentiate both side of (4) respect to θ, we have

$$\int \widehat{\alpha}\frac{\partial p(x;\theta)}{\partial\theta}dx = \frac{\partial g(\theta)}{\partial\theta} \tag{5}$$

or

$$\int \widehat{\alpha} \frac{\partial ln\ p(x;\theta)}{\partial \theta} p(x;\theta) dx = \frac{\partial g(\theta)}{\partial \theta} \tag{6}$$

From the regularity condition, we can modify (6) as

$$\int (\widehat{\alpha} - \alpha) \frac{\partial ln\ p(x;\theta)}{\partial \theta} p(x;\theta) dx = \frac{\partial g(\theta)}{\partial \theta}. \tag{7}$$

We apply the Cauchy - Schwarz inequality in the form

$$\left(\int w(x) g(x) h(x) dx \right)^2 \leq \int w(x) g^2(x) dx \int w(x) h^2(x) dx, \tag{8}$$

where $w(x) \geq 0$. The equality occurs if and only if $g(x) = ch(x)$ for some constants c.
Let $w(x) = p(x;\theta)$, $g(x) = \alpha - \widehat{\alpha}$ and $h(x) = \dfrac{\partial ln\ p(x;\theta)}{\partial \theta}$.
Apply Cauchy Schwarz (8) inequality,

$$\left(\frac{\partial g(\theta)}{\partial \theta} \right)^2 \leq \int (\alpha - \widehat{\alpha})^2 p(x;\theta) dx \int \left(\frac{\partial ln\ p(x;\theta)}{\partial \theta} \right)^2 p(x;\theta) dx \tag{9}$$

We can rewrite (9) as

$$\mathrm{Var}(\widehat{\alpha}) \geq \frac{\left(\frac{\partial g(\theta)}{\partial \theta} \right)^2}{E\left[\left(\frac{\partial ln\ p(x;\theta)}{\partial \theta} \right)^2 \right]} \tag{10}$$

We mention the regularity condition

$$\int \frac{\partial ln\ p(x;\theta)}{\partial \theta} p(x;\theta) = 0. \tag{11}$$

Differentiate left side of (11) with respect to θ

$$\begin{aligned} \frac{\partial}{\partial \theta} \int \frac{\partial ln\ p(x;\theta)}{\partial \theta} p(x;\theta) &= \int \frac{\partial^2 ln\ p(x;\theta)}{\partial \theta^2} p(x;\theta) + \frac{\partial ln\ p(x;\theta)}{\partial \theta} \frac{\partial p(x;\theta)}{\partial \theta} dx \\ &= \int \frac{\partial^2 ln\ p(x;\theta)}{\partial \theta^2} p(x;\theta) + \left(\frac{\partial ln\ p(x;\theta)}{\partial \theta} \right)^2 p(x;\theta) dx \\ &= E\left(\frac{\partial^2 ln\ p(x;\theta)}{\partial \theta^2} \right) + E\left(\left(\frac{\partial ln\ p(x;\theta)}{\partial \theta} \right)^2 \right) \end{aligned}$$

With the right side of regularity condition always zero, then

$$-E\left(\frac{\partial^2 ln\ p(x;\theta)}{\partial \theta^2} \right) = E\left(\left(\frac{\partial ln\ p(x;\theta)}{\partial \theta} \right)^2 \right) \tag{12}$$

Replace (12) into inequality (10), we obtain

$$\mathrm{Var}(\widehat{\alpha}) \geq \frac{\left(\frac{\partial g(\theta)}{\partial \theta} \right)^2}{-E\left(\frac{\partial^2 ln\ p(x;\theta)}{\partial \theta^2} \right)}. \tag{13}$$

The inequality (2) is derived by setting $g(\theta) = \theta$ in (13).
The condition for equality is

$$\frac{\partial ln\ p(x;\theta)}{\partial \theta} = \frac{1}{c(\theta)} (\widehat{\theta} - \theta). \tag{14}$$

Differentiate both side of (14) and take the expectation,

$$-E\left(\frac{\partial^2 ln\ p(x;\theta)}{\partial\theta^2}\right) = \frac{1}{c(\theta)} \tag{15}$$

or

$$c(\theta) = \frac{1}{-E\left(\frac{\partial^2 ln\ p(x;\theta)}{\partial\theta^2}\right)} = I(\theta), \tag{16}$$

which gives (3). □

Remark 1. *The expectation from (1) and (2) is determined as*

$$E\left[\frac{\partial ln\ p(x,\theta)}{\partial\theta}\right] = \int \frac{\partial ln\ p(x,\theta)}{\partial\theta} p(x;\theta)dx,$$

and

$$E\left[\frac{\partial^2 ln\ p(x,\theta)}{\partial\theta^2}\right] = \int \frac{\partial^2 ln\ p(x,\theta)}{\partial\theta^2} p(x;\theta)dx.$$

2 Vector version

Theorem 2. *It is assumed that the PDF $p(x;\theta)$ satisfies the "regularity" conditions*

$$E\left[\frac{\partial ln\ p(x,\theta)}{\partial\theta}\right] = 0,\ \forall\theta, \tag{17}$$

where the expectation is taken with respect to $p(x;\theta)$. Then, the covariance matrix of any unbiased estimator $\widehat{\theta}$ satisfies

$$C_{\widehat{\theta}} - I^{-1}(\theta) \geq 0, \tag{18}$$

where ≥ 0 means that the matrix is positive semidefinite. The Fisher information matrix

$$[I(\theta)]_{ij} = -E\left[\frac{\partial^2 ln\ p(x;\theta)}{\partial\theta_i\theta_j}\right]$$

where the derivative are evaluated at the true value of θ and the expectation is taken with respect to $p(x;\theta)$. Furthermore, an unbiased estimator that makes $C_{\widehat{\theta}} = I^{-1}(\theta)$ if and only if

$$\frac{\partial ln\ p(x;\theta)}{\partial\theta} = I(\theta)(g(x) - \theta), \tag{19}$$

for some p-dimensional function g and some $p \times p$ matrix I. $\widehat{\theta} = g(x)$ is MVU estimator and its covariance matrix is $I^{-1}(\theta)$.

Proof. □

3 Some example for scalar CRLB

References

[Kay] *Fundamentals of Statistical Signal Processing, Volume 1: Estimation Theory*. Pearson Education.